

$H_1(z)$ and $H_2(z)$ have GLP

- We can relax this to *generalized linear phase*:

$$H(e^{j\omega}) = A(\omega) e^{-j(\omega\ell + \phi_0)} \text{ for } 2\ell \in \mathbb{Z} \text{ and } \phi_0 \in \{0, \frac{\pi}{2}\}$$

while still retaining the desirable phase behavior.

Q: Does $H(z) = H_1(z)H_2(z)$ have GLP?

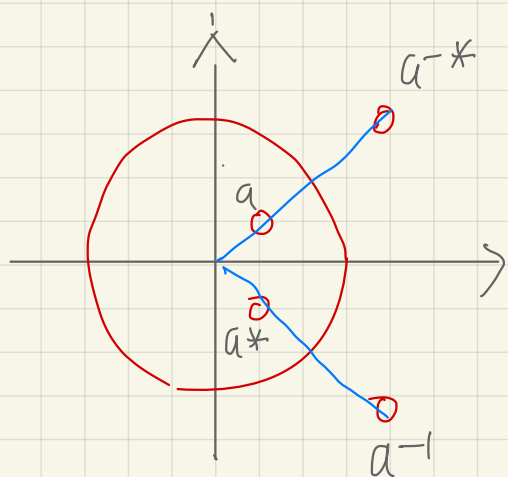
$$H_1(e^{j\omega}) = A_1(\omega) e^{-j(\omega\ell_1 + \phi_1)} \quad 2\ell_1, 2\ell_2 \in \mathbb{Z}$$

$$H_2(e^{j\omega}) = A_2(\omega) e^{-j(\omega\ell_2 + \phi_2)} \quad \phi_1, \phi_2 \in \{0, \frac{\pi}{2}\}$$

$$H_1(e^{j\omega}) H_2(e^{j\omega}) = A_1(\omega) A_2(\omega) e^{-j(\omega(\ell_1 + \ell_2) + \phi_1 + \phi_2)}$$

$$2\ell_1 + 2\ell_2 \in \mathbb{Z}$$

$$\begin{array}{l} \phi_1 + \phi_2 \\ 0 \\ \pi/2 \\ \pi \end{array}$$



reciprocal zeros

$$0 < r < 1$$

$$\theta \notin \{0, \pi\}$$

$$a = r e^{j\theta}$$

$$a^* = r e^{-j\theta}$$

$$a^{-*} = r^{-1} e^{j\theta}$$

$$a^{-1} = r^{-1} e^{-j\theta}$$

$$\begin{aligned} H(z) &= (z-a)(z-a^*)(z-a^{-*})(z-a^{-1}) \\ &= (z-a)(z-a^{-1})(z-a^*)(z-a^{-*}) \end{aligned}$$

Does $H(z)$ have GLP?

$$H(z) = (z^2 - (a+a^*)z + |a|^2) (z^2 - (a^{-1}+a^{-*})z + |a|^{-2})$$

$$= z^4 - (a+a^*+a^{-1}+a^{-*})z^3$$

$$+ [|a|^2 + |a|^{-2} - (2 + aa^{-*} + a^*a^{-1})] z^2$$

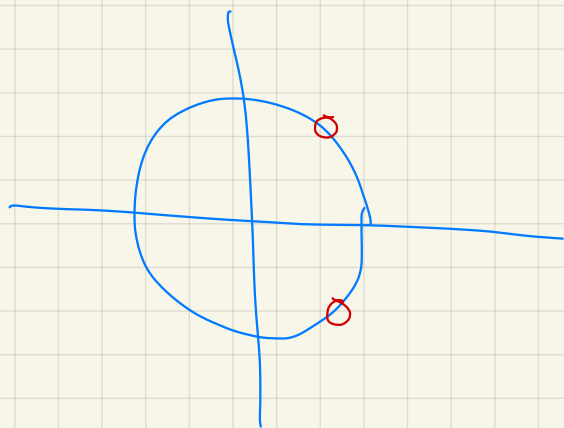
$$- [|a|^2(a^{-1}+a^{-*}) + |a|^{-2}(a+a^*)] z$$

$$+ |a|^2 |a|^{-2}$$

$$|a|^2 a^{-1} + |a|^2 a^{-*} + |a|^{-2} a + |a|^{-2} a^*$$

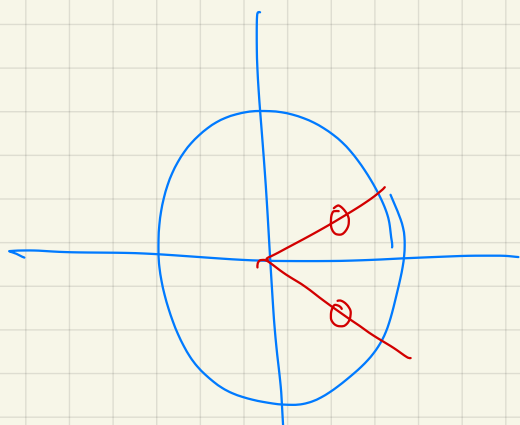
$$= r e^{-j\theta} + r e^{j\theta} + r^{-1} e^{j\theta} + r^{-1} e^{-j\theta}$$

$$= a^* + a + a^{-*} + a^{-1}$$



$$(z - a)(z - a^*) \quad |a| = 1$$

$$= z^2 - (a + a^*)z + |a|^2$$

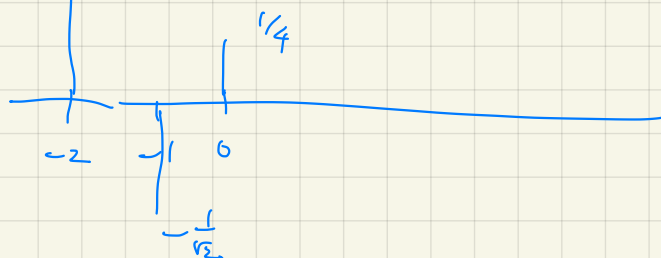


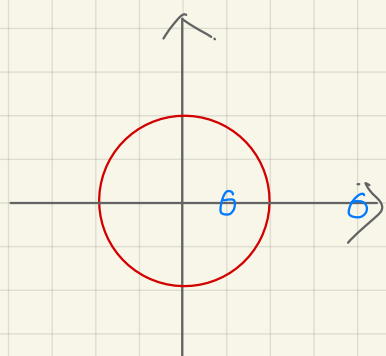
$$(z - \frac{1}{2}e^{j\frac{\pi}{4}})(z - \frac{1}{2}e^{-j\frac{\pi}{4}})$$

$$= z^2 - \underbrace{\frac{1}{2}(e^{j\frac{\pi}{4}} + e^{-j\frac{\pi}{4}})}_{\frac{1}{\sqrt{2}}} z + \frac{1}{4}$$

↕ inv $z + t/m$.

$$h[n] = \delta[n+2] - \frac{1}{\sqrt{2}}\delta[n+1] + \frac{1}{4}\delta[n]$$





reciprocal Zeros

$$a = r e^{j\theta}$$

$$0 < r < 1$$

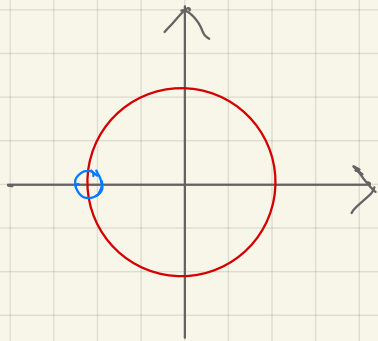
$$a^{-1} = \frac{1}{r} e^{-j\theta}$$

$$\theta \in [0, \pi]$$

$$H(z) = (z - a)(z - a^{-1}) = z^2 - (a + a^{-1})z + 1$$

Does $H(z)$ have GLP?

reciprocal zeros



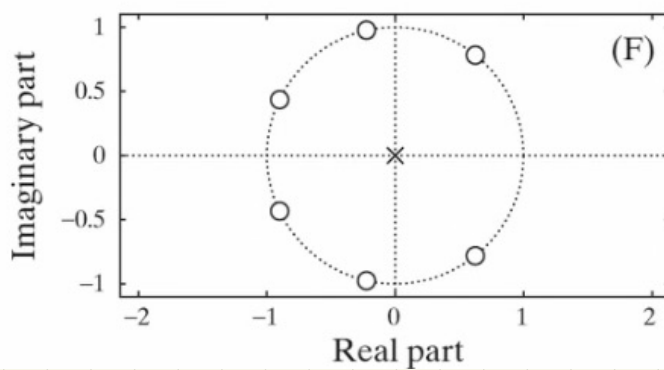
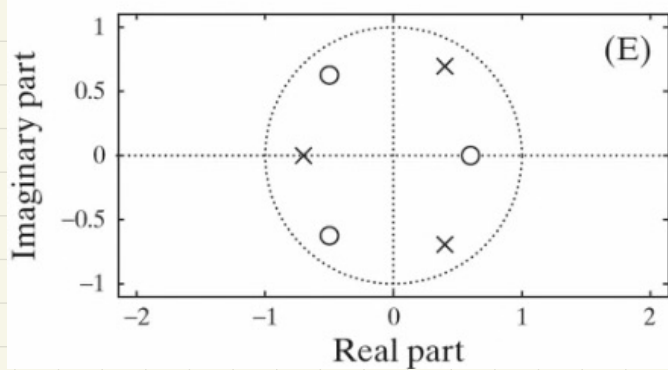
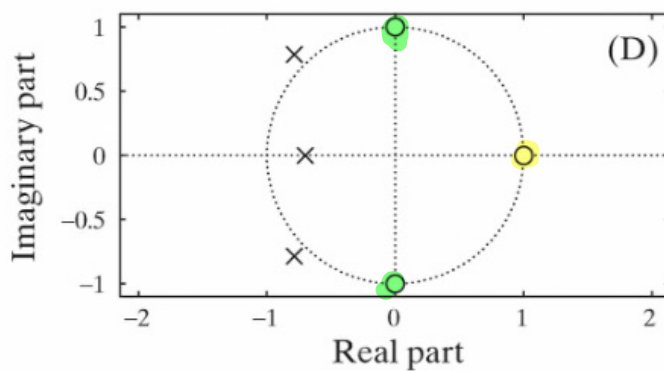
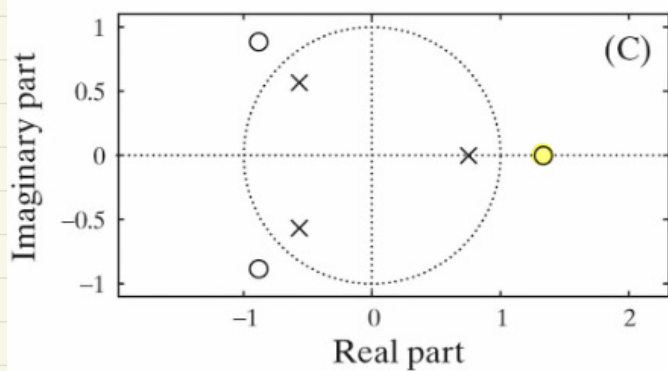
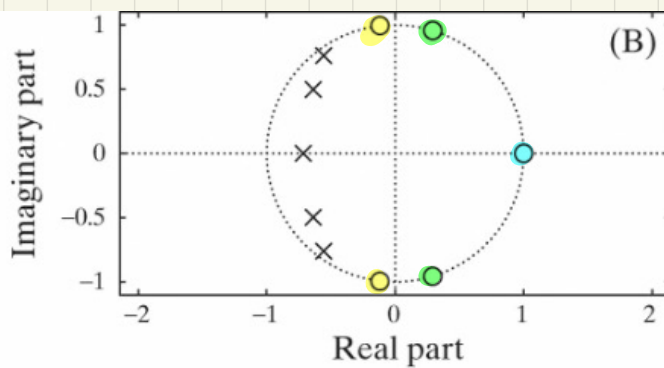
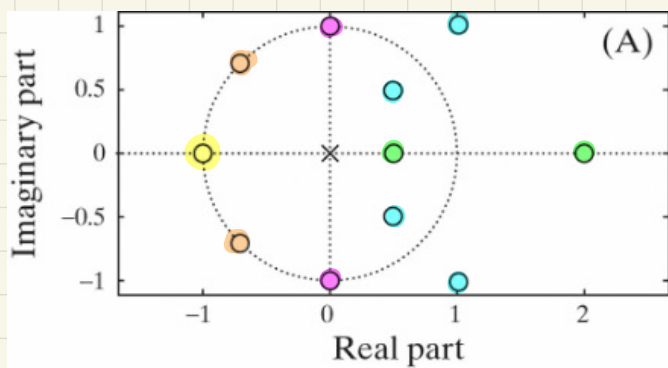
$$a = re^{j\theta}$$

$$H(z) = (z - a)$$

$$r = 1$$

$$\theta \in [0, \pi)$$

Does $H(z)$ have GLP?



IIR or FIR ?

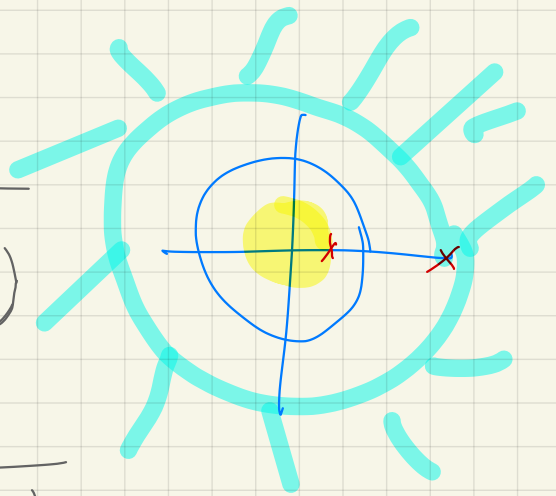
Assume causality. Which systems are stable?

GLP ?

lowpass ?

Q: Is there any IIR filter with GLP?

$$H(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}$$



$$H(e^{j\omega}) = \frac{1}{(1 - \frac{1}{2}e^{-j\omega})(1 - 2e^{-j\omega})}$$

$$= \frac{1}{1 - \frac{5}{2}e^{-j\omega} + e^{-j2\omega}}$$

$$= \frac{1}{e^{-j\omega}(e^{j\omega} + e^{-j\omega}) - \frac{5}{2}e^{-j\omega}}$$

$$= \frac{1}{2\cos\omega e^{-j\omega} - \frac{5}{2}e^{-j\omega}}$$

$$= \frac{e^{j\omega}}{2\cos\omega - \frac{5}{2}} = e^{j\omega} \cdot \frac{1}{2\cos\omega - \frac{5}{2}}$$

IIR filters:

- Recall that we have two ways to specify the transfer function of an IIR (i.e., recursive) filter:

1. A ratio of two polynomials in z^{-1} :

$$H(z) = \frac{\sum_{k=0}^N b_k z^{-k}}{1 + \sum_{m=1}^M a_m z^{-m}}$$

2. A ratio of two factored polynomials in z^{-1} :

$$H(z) = \frac{b_0 \prod_{k=1}^N (1 - z_k z^{-1})}{\prod_{m=1}^M (1 - p_m z^{-1})} \text{ with } \begin{cases} \text{zeros } \{z_k\}_{k=1}^N \\ \text{poles } \{p_m\}_{m=1}^M \end{cases}$$

In either case, the “order” = $\max(M, N)$.

- $H(z)$ is (*BIBO*) *stable* iff all poles are inside the unit circle: $|p_m| < 1$ for all m .
- Also recall that the poles and zeros come in complex-conjugate pairs (and/or real-valued singletons) iff $\{b_k\}$ and $\{a_m\}$ are real-valued.
- Next we’ll talk about ways of designing the filter coefficients. . .

$$s = \alpha \frac{z-1}{z+1}$$

$$s z + s = \alpha z - \alpha$$

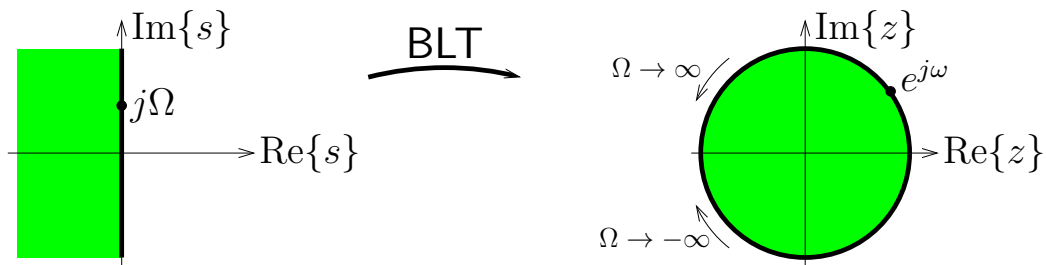
$$z = \frac{\alpha + s}{\alpha - s}$$

Bilinear transform:

$$s = j\Omega \Rightarrow z = \left| \frac{\alpha + j\Omega}{\alpha - j\Omega} \right| = \frac{\sqrt{\alpha^2 + \Omega^2}}{\sqrt{\alpha^2 + \Omega^2}} = 1$$

- One method of designing IIR digital filters is to start with an analog filter and “map” it to the digital domain.
- The *bilinear transform* accomplishes this mapping:

$$H(z) = H_c(s) \Big|_{s=\alpha \frac{z-1}{z+1}} = \alpha \frac{1-z^{-1}}{1+z^{-1}}$$



- The s -plane stability region/boundary is mapped to the z -plane stability region/boundary. $e^{j\omega} = \frac{\alpha + j\Omega}{\alpha - j\Omega}$
- The point $s = j\Omega$ is mapped to $z = e^{j\omega}$, where

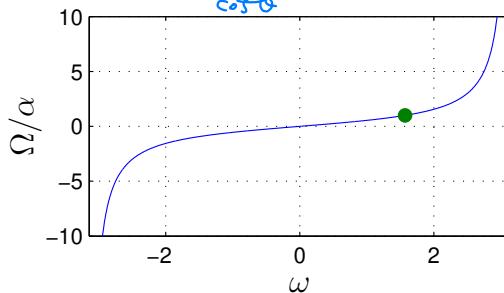
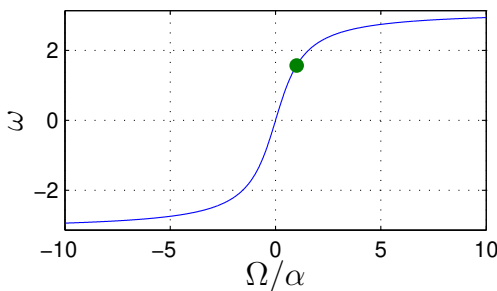
$$\omega = 2 \tan^{-1} \frac{\Omega}{\alpha} \quad (\text{e.g., } \Omega = \alpha \leftrightarrow \omega = \frac{\pi}{2})$$

$$\Omega = \alpha \tan \frac{\omega}{2} \quad (\text{e.g., } \omega = \pi \leftrightarrow \Omega = \infty)$$

$$\frac{e^{j\omega_1}}{e^{j\omega_2}} = e^{j(\omega_1 - \omega_2)}$$

$$\alpha + j\Omega = \sqrt{\alpha^2 + \Omega^2} \cdot \left(\frac{\alpha}{\sqrt{\alpha^2 + \Omega^2}} + j \frac{\Omega}{\sqrt{\alpha^2 + \Omega^2}} \right) = \sqrt{\alpha^2 + \Omega^2} \cdot (\cos \theta + j \sin \theta)$$

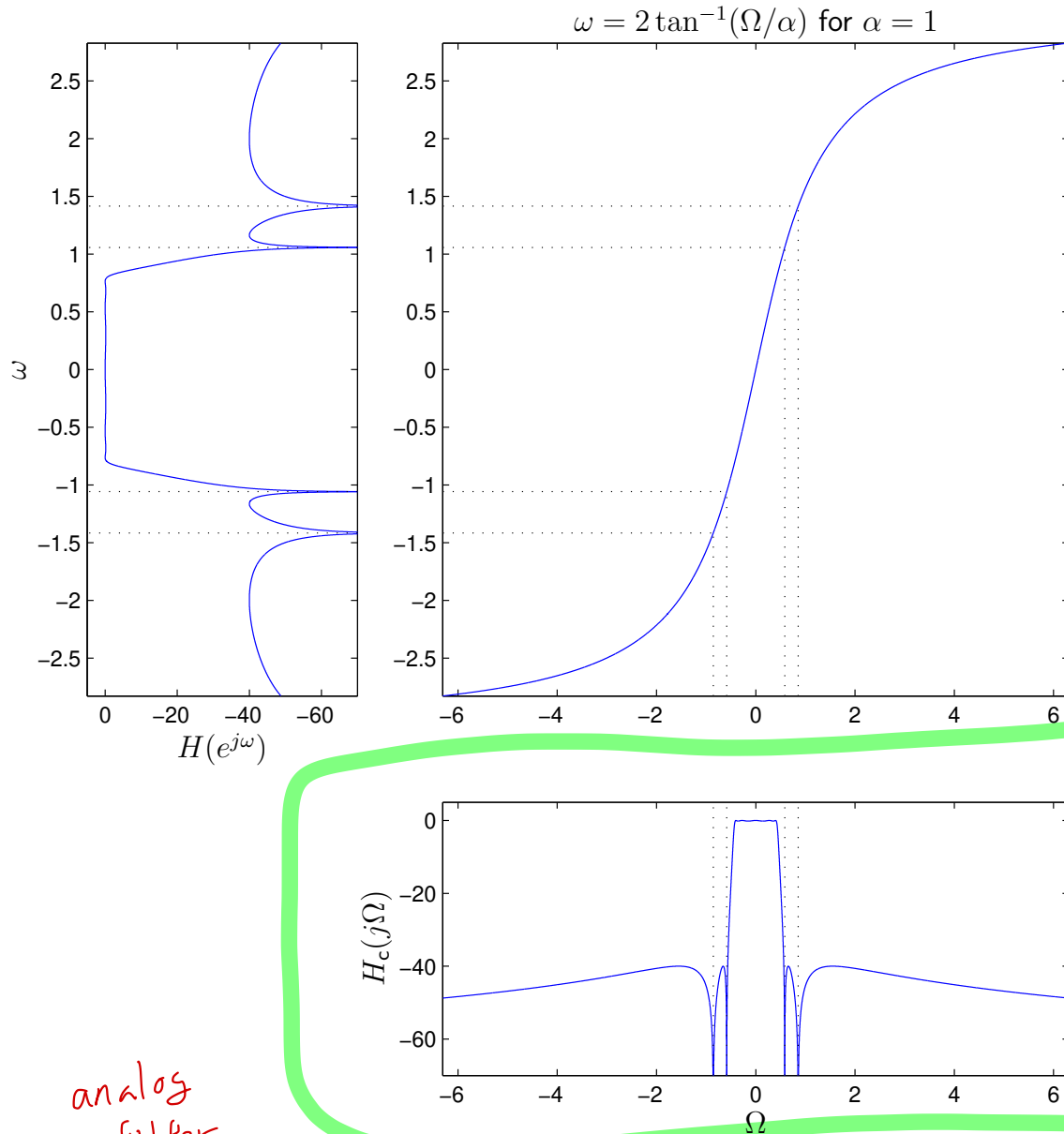
$$\theta = \tan^{-1} \left(\frac{\Omega}{\alpha} \right)$$



As we will see, the choice of α is arbitrary.

Bilinear transform (cont.):

Mapping a LPF from Ω -domain to ω -domain:



continuous-time analog filter

- Say ω_c is desired cutoff freq of $H(e^{j\omega})$. Then $H_c(j\Omega)$ is designed with cutoff frequency $\Omega_c = \alpha \tan \frac{\omega_c}{2}$. (Since we can accommodate any Ω_c , the choice of α doesn't matter!)
- Ripple in $H(e^{j\omega})$ will be identical to ripple in $H_c(j\Omega)$.

Analog prototypes:

- The most common analog prototype filters are:

filter type	passband	stopband	matlab	BLT?
Butterworth	smooth	smooth	butter	Y
Chebyshev	equiripple	smooth	cheby1	Y
	smooth	equiripple	cheby2	Y
Cauer (elliptic)	equiripple	equiripple	ellip	Y
Bessel	smooth	smooth	besself	N

- The above Matlab commands usually include the bilinear transform step, so you just specify the order N , desired cutoff ω_c , and filter type (LPF, HPF, etc.), and—in some cases—the ripple specs. (Note: For BPFs, N is doubled!)
- The corresponding commands `buttord`, `cheb1ord`, `cheb2ord`, and `ellipord` guess the required filter order.
- Note: Bessel analog filters have approximately linear phase through their passband. (Good!) However, they are LPF-only, have a wide transition band, and you have to perform manual BLT from the `besself` output, i.e.,

$$H(z) = \left. \frac{B_0 s^N + B_1 s^{N-1} + \dots + B_N}{s^N + A_1 s^{N-1} + \dots + A_N} \right|_{s=\alpha \frac{z-1}{z+1}}$$

for the $\{B_n\}$ and $\{A_n\}$ returned by `besself`.

Butterworth

$$|H(j\Omega)|^2 = \frac{1}{1 + (j\Omega/j\Omega_c)^{2N}}$$

Chebyshev
type 2

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 V_N^2(\Omega/\Omega_c)}$$

$$V_N(x) = \cos(N \cos^{-1} x)$$

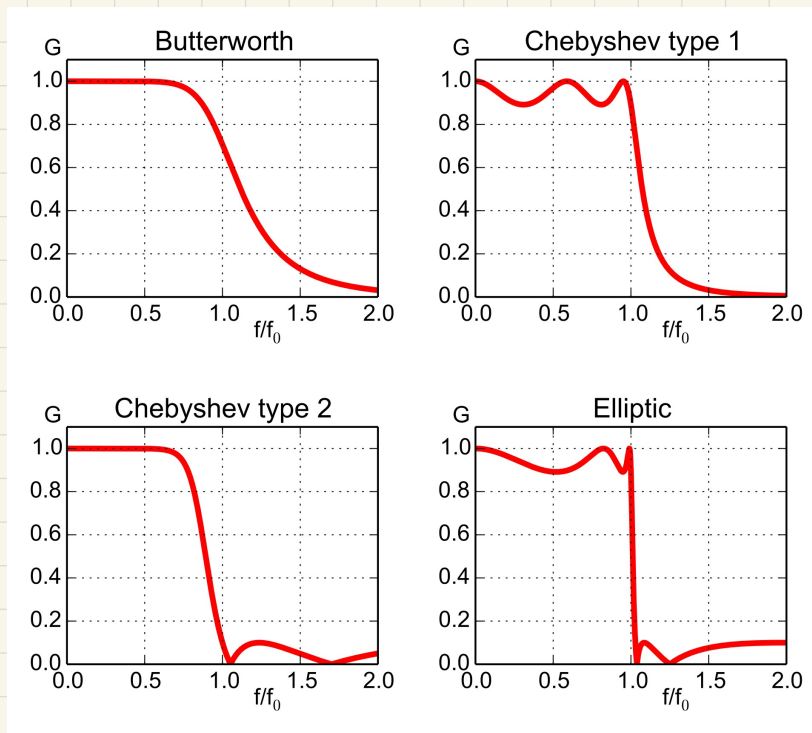
type 2

$$|H(j\Omega)|^2 = \frac{1}{1 + [\epsilon^2 V_N^2(\Omega/\Omega_c)]^{-1}}$$

Elliptic

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 U_N^2(\Omega)}$$

$U_N(\Omega)$: Jacobian elliptic function



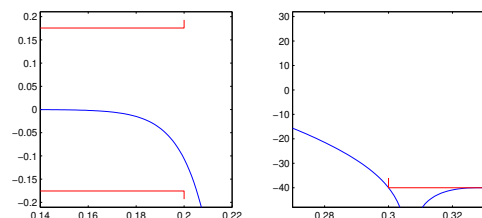
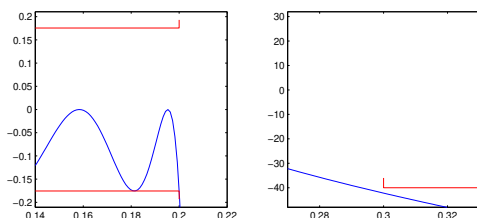
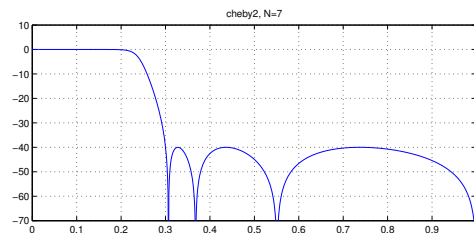
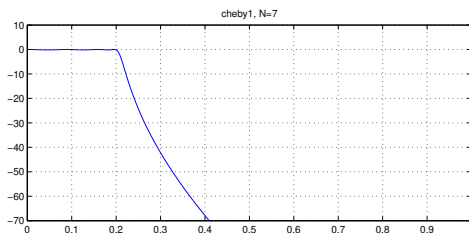
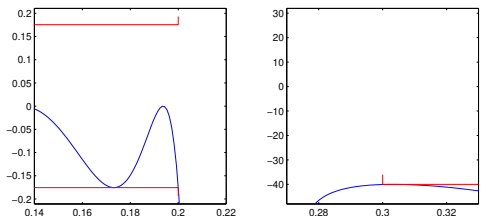
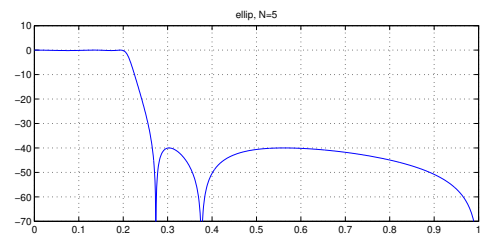
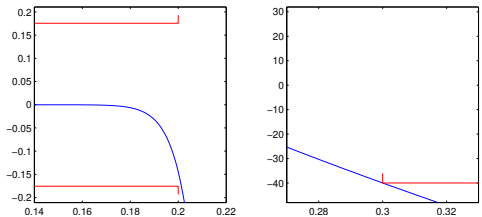
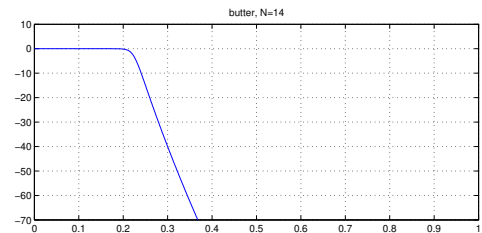
Analog-to-IIR examples:

```
fp = 0.2; % matlab-normalized frequency!
fs = 0.3;
delp = 0.02;
dels = 0.01;
```

```
Ap = -20*log10(1-delp);
As = -20*log10(dels);
```

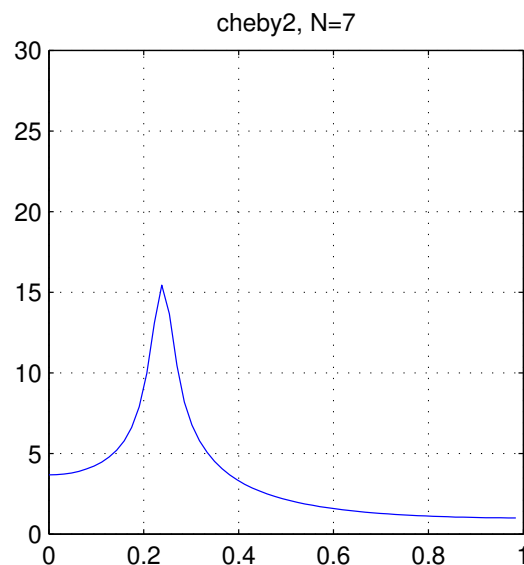
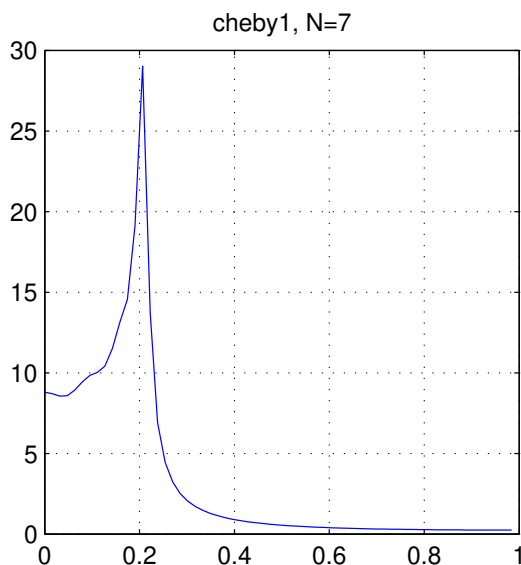
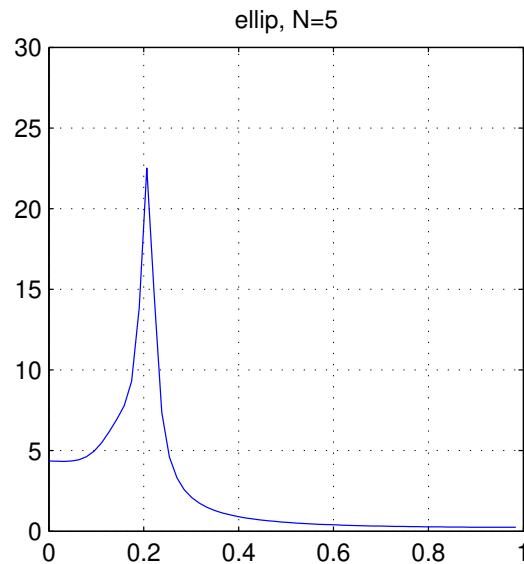
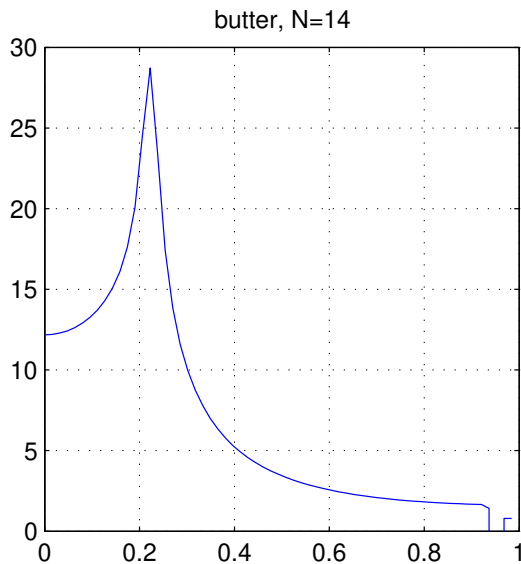
```
%[N,fc] = buttord(fp,fs,Ap,As); [b,a] = butter(N,fc);
%[N,fc] = cheb1ord(fp,fs,Ap,As); [b,a] = cheby1(N,Ap,fc);
%[N,fc] = cheb2ord(fp,fs,Ap,As); [b,a] = cheby2(N,As,fc);
[N,fc] = ellipord(fp,fs,Ap,As); [b,a] = ellip(N,Ap,As,fc)
```

```
[H,omega]=freqz(b,a,1024);
f = omega/pi;
subplot(2,1,1);
    plot(f,20*log10(abs(H)));
    grid on;
    axis([0,1,-70,10])
subplot(2,2,3);
    plot(f,20*log10(abs(H)),...
        [0,fp,fp],-Ap*[1,1,1.1], 'r',...
        [0,fp,fp],Ap*[1,1,1.1], 'r');
    axis([0.7*fp,1.1*fp,-1.2*Ap,1.2*Ap]);
subplot(2,2,4);
    plot(f,20*log10(abs(H)),...
        [fs,fs,1],-As*[0.9,1,1], 'r');
    axis([0.9*fs,1.1*fs,-1.2*As,0.8*As]);
```



Analog-to-IIR examples (cont.):

Group delay of the filters on the previous page:



The group delay in the passband (which is the region we care about most) is far from flat!

Padé's IIR method:

Consider design of the causal IIR filter:

$$H(z) = \frac{\sum_{k=0}^N b_k z^{-k}}{1 + \sum_{m=1}^M a_m z^{-m}}$$

Since

$$H(z) = \sum_{k=0}^N b_k z^{-k} - \sum_{m=1}^M a_m z^{-m} H(z),$$

we have

$$\begin{aligned} h[n] &= \sum_{k=0}^N b_k \delta[n-k] - \sum_{m=1}^M a_m h[n-m] \\ &= \begin{cases} 0 & n < 0 \\ b_n - \sum_{m=1}^M a_m h[n-m] & 0 \leq n \leq N \\ -\sum_{m=1}^M a_m h[n-m] & N < n \end{cases} \end{aligned}$$

causal

Main idea: We can *exactly match* $\{h[n]\}$ to a desired causal impulse response $\{d[n]\}$ at indices $n = 0 \dots N+M$! It is done in two steps:

1. Solve for $\{a_m\}_{m=1}^M$ that yield a match at $n = N+1 \dots N+M$:

$$\underbrace{- \begin{bmatrix} d[N+1] \\ \vdots \\ d[N+M] \end{bmatrix}}_{\mathbf{d}_{\text{padé}}} = \underbrace{\begin{bmatrix} d[N] & d[N-1] & \dots & d[N-M+1] \\ \vdots & \vdots & \ddots & \vdots \\ d[N+M-1] & d[N+M-2] & \dots & d[N] \end{bmatrix}}_{\mathbf{D}_{\text{padé}}} \underbrace{\begin{bmatrix} a_1 \\ \vdots \\ a_M \end{bmatrix}}_{\mathbf{a}}$$

$$\Rightarrow \mathbf{a}_{\text{padé}} = \mathbf{D}_{\text{padé}}^{-1} \mathbf{d}_{\text{padé}}$$

2. Solve for $\{b_n\}_{n=0}^N$ that yield a match at $n = 0 \dots N$:

$$b_n = d[n] + \sum_{m=1}^M a_{\text{padé},m} d[n-m] \text{ for } n = 0 \dots N.$$

Caveat: We are not even attempting to match $\{d[n]\}$ for $n > N+M$, so the designed response $\{h[n]\}$ might be really bad there!

Prony's method:

To prevent $\{h[n]\}$ and $\{d[n]\}$ from differing wildly when $n > N + M$, we can give up on *exactly* matching $\{d[n]\}_{n=0}^{N+M}$ and instead design $\{a_m\}_{m=1}^N$ to *approximately* match $\{d[n]\}_{n=N+1}^Q$ for some $Q > N + M$.

If we choose Q large enough, then usually both $d[n] \approx 0$ and $h[n] \approx 0$ for $n > Q$, giving the chance that $d[n] \approx h[n]$ everywhere.

For the approximate match, we can use a least-squares fit in place of the equality used for Padé's method (where $Q = N + M$):

$$\begin{aligned} \mathbf{a}_{\text{prony}} &= \underset{a_1 \dots a_M}{\operatorname{argmin}} \sum_{n=N+1}^Q \left| (-d[n]) - \sum_{m=1}^M d[n-m] a_m \right|^2 \\ &= \underset{\mathbf{a}}{\operatorname{argmin}} \left\| \underbrace{\begin{bmatrix} d[N] & \dots & d[N-M+1] \\ d[N+1] & \dots & d[N-M+2] \\ \vdots & & \vdots \\ d[Q-1] & \dots & d[Q-M] \end{bmatrix}}_{\mathbf{D}_{\text{prony}}} \begin{bmatrix} a_1 \\ \vdots \\ a_M \end{bmatrix} - \underbrace{\begin{bmatrix} -d[N+1] \\ -d[N+2] \\ \vdots \\ -d[Q] \end{bmatrix}}_{\mathbf{d}_{\text{prony}}} \right\|^2 \\ &= (\mathbf{D}_{\text{prony}}^H \mathbf{D}_{\text{prony}})^{-1} \mathbf{D}_{\text{prony}}^H \mathbf{d}_{\text{prony}} \end{aligned}$$

We then solve for $\{b_n\}_{n=0}^N$ as we did in Padé's method:

$$b_n = d[n] + \sum_{m=1}^M a_{\text{prony},m} d[n-m] \quad \text{for } n = 0 \dots N.$$

In summary, Prony's method gives

$$h[n] = d[n] \quad \text{for } n = 0 \dots N$$

$$h[n] \approx d[n] \quad \text{for } n = N+1 \dots Q$$

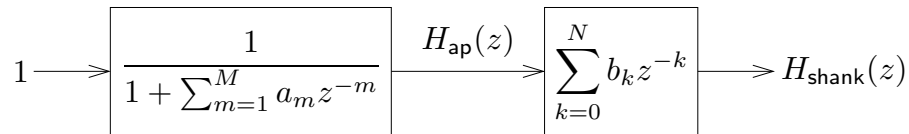
and hopefully $h[n] \approx 0 \approx d[n]$ for $n > Q$.

Shank's IIR method:

Shank's method is an improvement on Prony's method that allows $\{b_k\}_{k=0}^N$, as well as $\{a_m\}_{m=1}^M$, to approximately fit the actual $\{h[n]\}$ to the desired $\{d[n]\}$ over $n = 0 \dots Q$, for some $Q > N + M$.

Shank's method takes a two-step approach:

1. design the causal all-pole $H_{ap}(z) = \frac{1}{1 + \sum_{m=1}^M a_m z^{-m}}$ that gives the best LS fit to desired impulse response $\{d[n]\}$ over $n = 1 \dots Q$, then
2. design $\{b_k\}_{k=0}^N$ to best improve $\{h_{ap}[n]\}$, again using LS. N+1



Step 1)

Similar to Prony (but fit $\{d[n]\}_{n=1}^Q$ rather than $\{d[n]\}_{n=N+1}^Q$):

$$\mathbf{a}_{\text{shank}} = \underset{\mathbf{a}}{\text{argmin}} \left\| \underbrace{\begin{bmatrix} d[0] & 0 & 0 \\ \vdots & \ddots & 0 \\ d[Q-M] & & d[0] \\ \vdots & \ddots & \vdots \\ d[Q-1] & \dots & d[Q-M] \end{bmatrix}}_{\mathbf{D}_{\text{shank}}} \begin{bmatrix} a_1 \\ \vdots \\ a_M \end{bmatrix} - \underbrace{\begin{bmatrix} -d[1] \\ \vdots \\ -d[Q-M+1] \\ \vdots \\ -d[Q] \end{bmatrix}}_{\mathbf{d}_{\text{shank}}} \right\|^2$$

$$= (\mathbf{D}_{\text{shank}}^H \mathbf{D}_{\text{shank}})^{-1} \mathbf{D}_{\text{shank}}^H \mathbf{d}_{\text{shank}}$$

This yields (from block diagram)

$$H_{ap}(z) = 1 - \sum_{m=1}^M a_{\text{shank},m} z^{-m} H_{ap}(z)$$

$$\Rightarrow h_{ap}[n] = \begin{cases} 0 & n < 0 \\ 1 & n = 0 \\ -\sum_{m=1}^M a_{\text{shank},m} h_{ap}[n-m] & n > 0 \end{cases}$$

Shank's IIR method (cont.):

Step 2)

From block diagram, we see that

$$H_{\text{shank}}(z) = \sum_{k=0}^N b_k z^{-k} H_{\text{ap}}(z)$$

$$\Rightarrow h_{\text{shank}}[n] = \sum_{k=0}^N b_k h_{\text{ap}}[n - k]$$

and so, to yield the best LS fit of $\{h_{\text{shank}}[n]\}$ to $\{d[n]\}$ over $n = 0 \dots Q$, we solve for

$$\begin{aligned} \mathbf{b}_{\text{shank}} &= \underset{b_0 \dots b_N}{\operatorname{argmin}} \sum_{n=0}^Q \left| d[n] - \sum_{k=0}^N b_k h_{\text{ap}}[n - k] \right|^2 \\ &= \underset{\mathbf{b}}{\operatorname{argmin}} \left\| \underbrace{\begin{bmatrix} h_{\text{ap}}[0] & 0 & 0 \\ \vdots & \ddots & 0 \\ h_{\text{ap}}[Q-N] & & h_{\text{ap}}[0] \\ \vdots & \ddots & \vdots \\ h_{\text{ap}}[Q] & \dots & h_{\text{ap}}[Q-N] \end{bmatrix}}_{\mathbf{H}_{\text{ap}}} \underbrace{\begin{bmatrix} b_0 \\ \vdots \\ b_N \end{bmatrix}}_{\mathbf{b}} - \underbrace{\begin{bmatrix} d[0] \\ \vdots \\ d[Q-N] \\ \vdots \\ d[Q] \end{bmatrix}}_{\mathbf{d}} \right\|^2 \\ &= (\mathbf{H}_{\text{ap}}^H \mathbf{H}_{\text{ap}})^{-1} \mathbf{H}_{\text{ap}}^H \mathbf{d} \end{aligned}$$